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Full Length Article

DPDA: A dual-path domain adaptation method for cross-terrain real-time attack detection in mobile robots

Yu Wang [a], Zhitao He [a], Yongyi Chen [a], Dan Zhang [a,*], Hamid Reza Karimi [b]

[a] Research Center of Automation and Artificial Intelligence, Zhejiang University of Technology, Hangzhou, 310023, Zhejiang, China
[b] Department of Mechanical Engineering, Politecnico di Milano, Milan, 20156, Italy

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ABSTRACT

Deep learning-based attack detection methods for mobile robots achieve high accuracy within a single operating environment, but exhibit significant limitations when deployed in real-world cross-terrain scenarios. The core issue stems from the inability of single-domain models to adapt to data distribution shift across terrains, resulting in the misclassification of terrain-induced variations as potential attacks. To overcome this challenge, a dual-path domain adaptation (DPDA) network is proposed for cross-terrain real-time attack detection in mobile robots. The temporal dynamics and spectral characteristics of sensor data are extracted through the dual-path architecture in DPDA, overcoming the informational constraints of unimodal approaches. Furthermore, a time-frequency correlator (TFC) and a bidirectional cross attention (BCA) mechanism are introduced to enable interactive fusion of bimodal features, capturing stable attack features across domains. To tackle the dual-target domain alignment problem, a reciprocal prototype drift (RPD) mechanism is designed to bridge disparate target domains. By guiding feature prototypes from different target domains to converge mutually in the embedding space, distribution shift induced by terrain variations is effectively mitigated. Experiments conducted on a real mobile robot platform demonstrate that DPDA surpasses existing methods in cross-terrain attack detection tasks. Cross-dataset validation further confirms the model's robust cross-domain generalization capability.

1. Introduction

Autonomous systems are becoming increasingly integral to modern intelligent transportation infrastructures. As mobile robots transition from structured urban roads to unstructured off-road environments, cross-terrain autonomous navigation has emerged as a critical capability across a wide range of applications (Guan et al., 2022; Miki et al., 2022; Weerakoon et al., 2022). The control and safety decisions of mobile robots rely on real-time sensor feedback which requires frequent data exchange, rendering the communication network vulnerable to malicious attacks (Ma et al., 2023a; Ueda & Kwon, 2025). Adversaries can manipulate transmitted sensor measurements to deviate the robot from its planned trajectory, posing serious threats to operational safety (Zhang et al., 2023). Variations across different terrains induce a significant shift in the statistics of sensor data, further complicating the accurate detection of anomalies. Consequently, the development of a real-time attack detection system with robust cross-terrain generalization capability is of paramount importance.

The core of anomaly detection lies in accurately distinguishing natural disturbances from malicious signals in time-series data. Supervised deep learning (DL) approaches, leveraging powerful feature extraction and temporal modeling, have become dominant (Keshk et al., 2023; Li et al., 2025). For example, Ji et al. (2022) introduced a proactive anomaly detection network to enhance detection performance for robot navigation in complex environments. By planning the path and fusing sensor signals to predict the future failure probability, robotic anomalous behaviors can be captured in real time. Azzalini et al. (2021) presented a novel minimally supervised approach for anomaly detection in mobile robots. By adopting the variational autoencoders architecture and a novel training method, a progress-based latent space can be induced for anomaly detection. He et al. (2025) proposed the temporal shuffle attention network (TSAN) for sensor attack detection in mobile robots. Temporal nature of the sensor features is enhanced by positional encoding, and a multilevel TSA module is employed so that the data features can be fully extracted. Through the extraction and modeling of data features, the aforementioned methods improve the

* Corresponding author.
E-mail addresses: 221124030325@zjut.edu.cn (Y. Wang), 201806060711@zjut.edu.cn (Z. He), chenyy@zjut.edu.cn (Y. Chen), danzhang@zjut.edu.cn (D. Zhang), hamidreza.karimi@polimi.it (H.R. Karimi).

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ability to detect anomalies in robot networks within a single operating environment.

However, conventional DL-based attack detection methods struggle to address the evolving threats in dynamic operating environments (Lampe & Meng, 2024; Layeghy et al., 2023). The limitation stems from the fundamental assumption in single-domain methods that training and testing data are independent and identically distributed, which no longer holds true in cross-terrain scenarios (Khoee et al., 2024). When deployed, supervised DL methods experience a significant decline in detection performance due to domain shift (Kheddar et al., 2023; Zhang & Zhao, 2021). Domain shift arises from the influence of varying terrain properties on robot dynamics, which leads detectors to misinterpret terrain-induced normal variations as attacks, thereby increasing false alarm rates. Moreover, recollecting and annotating large-scale datasets for each new terrain is both economically costly and poorly scalable (Hoyer et al., 2023; Li et al., 2023). To mitigate performance degradation caused by the mismatch between training and testing environments, domain adaptation techniques emerge as a promising direction for enhancing attack detection capabilities across diverse environments.

Domain adaptation is achieved by establishing shared feature representations between source and target domains, thereby minimizing distribution discrepancies (Shi et al., 2022; Wu et al., 2022). Thus, the target domain can maintain high accuracy without extensive target labeling. Marchiori et al. (2025) proposed the so-called J-DAPT, a method for detecting jailbreaking attacks in robot attention-based fusion and domain adaptation. The core lies in utilizing domain adaptation to align general-purpose jailbreak datasets with domain-specific reference data, thereby addressing the issue where detectors often misclassify robot jailbreaking attacks as safe commands. Hu et al. (2022) introduced a deep subdomain adaptation network with attention mechanism to identify malware variant traffic at an internet of things (IoT) edge gateway. Local maxsensorm mean discrepancy is utilized to align the traffic feature distributions of subdomains. Ma et al. (2023b) designed a few-shot adversarial discrimination domain adaptation framework for IoT intrusion detection. Dual domain hybrid and concatenation algorithms are adopted to alleviate over-fitting under few-shot constraints. Nevertheless, existing methods are predominantly tailored for images or static data and overlook the temporal dependencies inherent in time-series data (Huang et al., 2023; Li et al., 2018; Wang et al., 2020). Due to the dynamic features and spectral information inherent in temporal data, mere distribution alignment tends to result in the loss of these intrinsic properties. Additionally, existing approaches perform only source-to-target alignment and ignore inter-target feature correlations when handling dual-target domain adaptation, preventing the model from reaching optimal performance. The above shortcomings reduce the effectiveness of domain-adaptive attack detection in cross-terrain scenarios for mobile robots.

To address this gap, this paper investigates a more practical yet challenging scenario called multi-target domain adaptation-based anomaly detection, where the model is no longer restricted to a single terrain. The proposed method aims to enhance model generalization across diverse terrains using labeled source-domain supervision. To the best of our knowledge, multi-target domain adaptation for anomaly detection has not yet been fully explored in the existing literature. Although domain adaptation techniques and their associated components have been extensively studied in other fields, directly transferring these methods to cross-terrain anomaly detection still faces two challenges, which are summarized as follows:

1) Attack-related features are entangled with terrain-specific features, making it difficult for models to learn stable cross-terrain attack representations.

2) Existing methods perform only unidirectional source-target alignment, which leads to unconstrained feature drift between dual-target domains.

To address the aforementioned challenges, a dual-path domain adaptation (DPDA) network is proposed in this paper. Cross-terrain generalizable attack representations are learned by exploiting shared attack patterns across different terrains. Meanwhile, for the dual-target domain scenario, inter-target relationships are incorporated into the feature learning process, enabling mutual constraints and collaborative evolution between target domains. The key components are specifically tailored to the unique challenges of cross-terrain attack detection, and such a design has not been previously explored in the field of anomaly detection for mobile robots. Notably, this study addresses known scenarios requiring adaptation, where unlabeled data from the target terrain can be collected in advance via low-cost methods such as low-speed mapping. As a result, without requiring target domain attack annotations, the proposed method can mitigate terrain variation interference and focus on anomaly pattern detection. The principal contributions of this paper are summarized as follows:

- A cross-terrain attack detector DPDA is introduced to address the misclassification of terrain-induced perturbations as attacks. The model is guided to focus on attack patterns coexisting across different terrains, thereby suppressing terrain disturbance. To achieve this, cross-modal time-frequency feature learning and inter-target alignment are adopted to capture terrain-invariant attack patterns.
- The prototype alignment in dual-target domains is redefined as a bidirectional constrained drift process, based on which the RPD mechanism is designed. By exploiting the feature correlations between target domains, features from one target domain are guided toward the prototypes of the other, alleviating unconstrained drift in dual-target alignment.
- To prevent spectral representations from being dominated by terrain-induced spectral variations, domain probabilities are employed as gating signals to mitigate domain-specific spectral components, while the remaining interference is further attenuated through bidirectional attention, promoting the extraction of stable attack features.

The remainder of this article is organized as follows. Section 2 details the proposed methodology. Section 3 evaluates the model with real mobile robot sensor data and performs some comparison studies. Section 4 analyzes our model. Section 5 summarizes the main work.

2. Proposed method

2.1. Overview

To address the data distribution shift encountered by mobile robots across terrains, a novel domain adaptation method is proposed, i.e., DPDA. The primary objective is to mitigate the performance degradation caused by transferring the model from one terrain to another. The core idea of DPDA is to jointly leverage temporal and spectral features through a dual-path architecture, combined with an adaptive domain alignment mechanism that facilitates cross-terrain generalization. As a result, the proposed model achieves robust anomaly detection even under significant terrain variations.

The overall framework of DPDA is illustrated in Fig. 1. The network takes as input labeled source samples X^s and unlabeled target samples X^{t1} and X^{t2}. Firstly, a parallel temporal convolutional network and a spectral branch extract discriminative representations from the raw time-series and their frequency spectra, respectively. Subsequently, the time-frequency correlator (TFC) module dynamically adjusts the contribution weights of time-frequency features according to the domain probabilities, thus mitigating domain-specific spectral responses. Next, a consistent representation of attack-related information is established by the bidirectional cross attention (BCA) mechanism. To reduce the distribution discrepancy between the source and target domains, GRL-

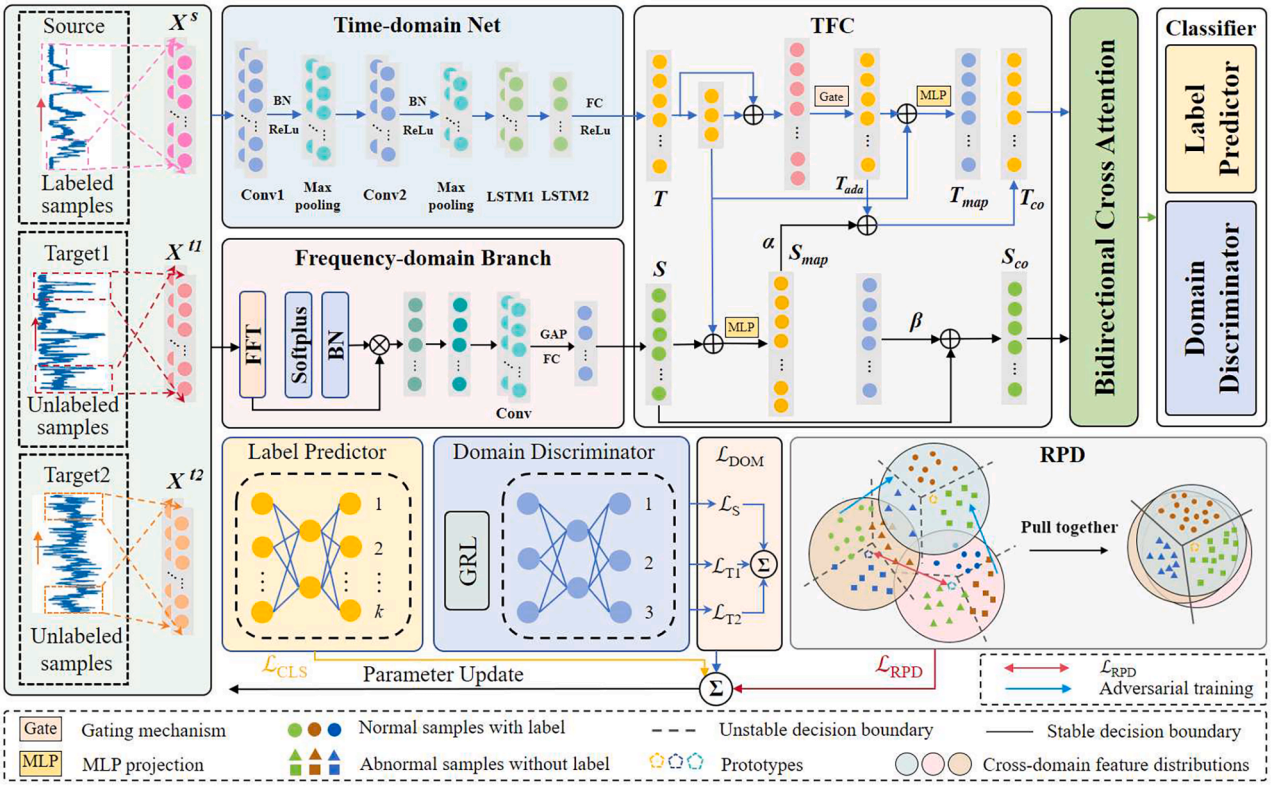


Fig. 1. Overall framework of the proposed DPDA method.

based adversarial domain adaptation is introduced, enabling the feature extractor to learn more domain-invariant representations through adversarial training. To mitigate the discrepancy between dual target domains, the RPD is introduced to encourage mutual guidance and collaborative optimization among different target domains in the feature space. Finally, the fused representation is passed to a classifier to achieve precise detection across diverse terrain scenarios.

2.2. Dual-path architecture

To overcome the representational limitation of single-modal approaches in detection task, the dual-path architecture is proposed to enhance feature representations by processing temporal and spectral information in parallel. The design is motivated by the observation that attack-related features in robotic sensor data manifest complex variation patterns across different terrains. Conventional single-modality feature extraction methods fail to adequately capture such complexity, particularly under significant distribution shift between source and target domains. Accordingly, a parallel backbone network is designed to extract discriminative features from complementary perspectives, thereby providing richer feature representations for subsequent domain adaptation. Detailed parameters of the time-domain network are listed in Table 1.

A CNN-LSTM architecture is adopted in the time-domain network to capture dynamic features of attack behaviors from raw sensor data. The local temporal correlations within the input signals X^s, X^{t1}, and X^{t2} are first captured by two layers of convolution. Batch normalization (BN) is then applied to mitigate initial shift across terrain data by standardizing feature distributions, followed by max-pooling to retain salient features while reducing dimensionality. Subsequently, long-short-range dependencies are modeled by a two-layer LSTM through its gating mechanism, capturing the dynamic characteristics of attack behaviors. The resulting time-domain feature vector T is forwarded to the TFC module as the temporal representation. In parallel, the amplitude spectrum of each input window is first computed by the spectral branch using fast Fourier transform. The different frequency components are then adaptively weighted by a trainable weight vector. The reweighted spectrum is up-projected and processed by a CNN to produce the frequency-domain feature vector S, which encodes terrain-invariant spectral structures. The dual-path design not only preserves the temporal dynamics of attack behavior but also captures terrain-related spectral information. As a result, richer and more complementary discriminative information is incorporated, enhancing the model's robustness to terrain-induced statistical variations.

Table 1
Time-domain network parameters in DPDA.

Layer(type)	Layer size	Channels	Output size
Input	–	1	(None, 100, 1)
Conv1/BN/ReLu	3×1	64	(None, 100, 64)
MaxPooling	–	64	(None, 50, 64)
Conv2/BN/ReLu	3×1	128	(None, 50, 128)
MaxPooling	–	128	(None, 25, 128)
LSTM1	–	128	(None, 25, 128)
LSTM2	–	64	(None, 64)
FC	–	64	(None, 64)

2.3. TFC module and BCA mechanism

The temporal and spectral features extracted are weakly correlated, causing spectral representations to be easily dominated by domain-specific frequency responses. Therefore, the classifier struggles to distinguish terrain-induced noise from actual attacks. To overcome this limitation, the TFC module is deployed to mitigate domain-specific spectral components, after which BCA performs deep correlation and fusion of features, further attenuating residual interference.

2.3.1. TFC

The core of TFC lies in introducing cross-modal interaction, enabling temporal and frequency domain features to achieve mutual perception and calibration prior to fusion. Let the temporal feature of the dual-

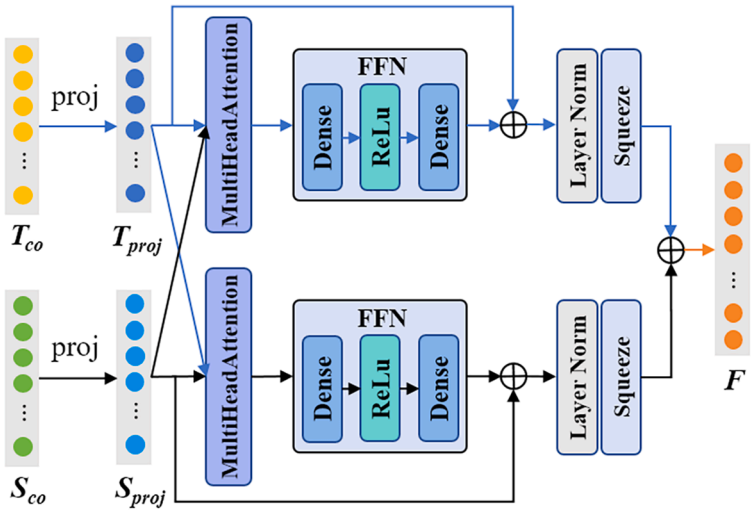


Fig. 2. Flow chart of the BCA mechanism.

path module output be $T \in \mathbb{R}^{B\times 64}$ (B represents the batch size, and 64 denotes the feature dimension), and the spectral feature be $S \in \mathbb{R}^{B\times 32}$. A domain attribution probability distribution is first computed via a fully connected (FC) layer:

$$p = \mathrm{Softmax}(W_d T + b_d) \in \mathbb{R}^{B\times 3} \tag{1}$$

where the three dimensions of p are assigned to the predictive probabilities for the source domain and the dual target domains, respectively. $W_d \in \mathbb{R}^{3\times 64}$ represents the domain discriminative weight matrix, and b_d denotes the bias term.

Subsequently, dynamic weights are generated through a gating mechanism.

$$g = \mathrm{Sigmoid}(W_g \hat{T} + b_g) \tag{2}$$

where $\hat{T} = [T, p] \in \mathbb{R}^{B\times 67}$ and $W_g \in \mathbb{R}^{B\times 32}$ are the gating weight matrices.

The final result is the adapted time-domain feature $T_{ada} = T \odot g$, $\odot$ represents element-wise multiplication. The above operation dynamically suppresses terrain-related noise components through domain probability, preserving the temporal attack feature baseline. Next, the features are projected across domains using a multilayer perceptron (MLP), mapping temporal features to the frequency dimension and vice versa.

$$T_{map} = \mathrm{ReLU}(W_{t2s}\hat{T} + b_{t2s}) \tag{3}$$

$$S_{map} = \mathrm{ReLU}(W_{s2t}\hat{S} + b_{s2t}) \tag{4}$$

where $\hat{T} = [S, p] \in \mathbb{R}^{B\times 35}$, $W_{t2s} \in \mathbb{R}^{32\times 67}$ and $W_{s2t} \in \mathbb{R}^{64\times 35}$ are the mapping weight matrices. Then learnable parameters are introduced to control their fusion weights, ultimately yielding the following collaborative features.

$$T_{co} = T_{ada} + \alpha \cdot S_{map} \tag{5}$$

$$S_{co} = S + \beta \cdot T_{map} \tag{6}$$

where $\alpha, \beta \in \mathbb{R}^{+}$ are learnable fusion parameters. The collaborative processing is designed to automatically adjust contribution weights based on terrain features in the temporal and frequency domains, thereby filtering out terrain-related information.

2.3.2. BCA

To enhance the stability of the feature space in cross-terrain scenarios, the BCA mechanism is designed to attenuate residual interference induced by terrain variations. Through bidirectional interaction modeling, the mechanism facilitates the fusion of bimodal features and ensures a consistent representation of attack-relevant information. The BCA mechanism flow chart is shown in Fig. 2. The cooperative features output by TFC are first passed through a projection layer to unify their dimensions.

$$T_{proj} = W_t T_{co} + b_t \tag{7}$$

$$S_{proj} = W_s S_{co} + b_s \tag{8}$$

where $W_t \in \mathbb{R}^{64\times 64}$ and $W_s \in \mathbb{R}^{64\times 32}$ are the projection matrix. To satisfy the sequence format of the attention mechanism, the features are expanded.

$$\tilde{T} = T_{proj} \otimes e_1 \tag{9}$$

$$\tilde{S} = S_{proj} \otimes e_1 \tag{10}$$

Subsequently, the temporal features are enhanced using the multi-head attention mechanism with frequency-domain context. Bidirectional attention is computed in two directions, temporal to frequency (Attn_t) and frequency to temporal (Attn_s).

$$\mathrm{Attn}_t = \mathrm{MultiHead}(\tilde{T}, \tilde{S}, \tilde{S}) \tag{11}$$

$$\mathrm{Attn}_s = \mathrm{MultiHead}(\tilde{S}, \tilde{T}, \tilde{T}) \tag{12}$$

For a single attention head, the computation is

$$\mathrm{head}_i = \mathrm{Softmax}\left(\frac{\left(\tilde{T}W_i^Q\right)\left(\tilde{S}W_i^K\right)^{\mathrm{T}}}{\sqrt{d_k}}\right)\left(\tilde{S}W_i^V\right) \tag{13}$$

where $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{64\times 16}$ are the parameter for the ith head. $d_k = 16$ is the head dimension.

Following the attention output, residual connection and feed-forward network (FFN) are employed to enhance effective features while suppressing redundant information.

$$T_{out} = \mathrm{LayerNorm}\left(\tilde{T} + \mathrm{FFN}\left(\mathrm{Attn}_t\right)\right) \tag{14}$$

$$S_{out} = \mathrm{LayerNorm}\left(\tilde{S} + \mathrm{FFN}\left(\mathrm{Attn}_s\right)\right) \tag{15}$$

where $\mathrm{FFN}(x) = W_2\mathrm{ReLU}(W_1 x + b_1) + b_2$. $W_1 \in \mathbb{R}^{64\times 64}$ and $W_2 \in \mathbb{R}^{64\times 64}$ are FFN parameters. The final fusion feature is

$$F = [\mathrm{squeeze}(T_{out}), \mathrm{squeeze}(S_{out})] \in \mathbb{R}^{B\times 128} \tag{16}$$

Through the attention mechanism, the core correlations of time-frequency features are weighed and selected, thereby yielding a cross-modal cooperative representation. The design effectively mitigates the interference of environmental noise on feature matching across various terrains, enabling the model to more readily capture stable features across domains.

Overall, terrain-correlated components in spectral feature are suppressed by the TFC module, and more stable input representations are

provided for subsequent domain adaptation. Meanwhile, the BCA mechanism further amplifies feature discriminability via dual-dimensional interactive modeling, accurately capturing anomaly patterns under domain shift. Operating in tandem, both components establish a robust feature foundation for domain adaptation attack detection across terrains.

2.4. Domain adaptation model

In cross-terrain attack detection for mobile robots, distribution shift in sensor data severely degrades model performance on other terrains. To resolve this challenge, we incorporate adversarial domain adaptation, which establishes an adversarial game between a domain discriminator and a feature extractor. By introducing a gradient reversal layer (GRL), the feature extractor is guided to generate domain-invariant feature representations that confuse the domain discriminator. The process suppresses terrain-specific variations in the learned features, thereby encouraging the model to concentrate on domain-agnostic patterns essential for attack detection. As a result, the discriminative capability of the feature representations is enhanced, leading to improved generalization across diverse terrain conditions.

Moreover, existing domain adaptation methods typically concentrate on unidirectional alignment between the source domain and a single target domain. For example, prototype alignment methods employ source-domain prototypes as anchors to guide target features toward corresponding source classes, whereas centroid matching methods reduce domain discrepancy by directly minimizing the distance between domain centroids (Li et al., 2022; Tian et al., 2020). Although these strategies are proved to be effective in single-target domain adaptation, they primarily enforce source-target consistency and do not explicitly account for the interrelationships among multiple target domains. When two target domains are present simultaneously, factors such as terrain-induced noise and optimization stochasticity may cause the two target domains to drift toward disparate regions in the feature space (Yang et al., 2022). This drift, in turn, leads to an unstable decision boundary and consequently degrades anomaly detection performance.

To address this issue, the RPD module is designed, as illustrated in Fig. 1. By establishing a dynamic bidirectional guidance mechanism, RPD enables the prototypes of the two target domains to mutually constrain and collaboratively update each other during the feature evolution process. The features of target domains are guided to drift toward each other's prototypes, which is achieved by computing and maintaining the feature prototypes of dual target domains. The drift is then constrained from diverging through adversarial training. Specifically, the feature prototypes in the dual target domains are updated by the RPD using the exponential moving average (EMA) strategy. A drift loss is then computed based on the cosine distance between the features of one target domain and the prototype of the other. Through adversarial training, the reciprocal alignment strategy effectively promotes the mixing and interaction of target domain features within the embedding space.

Domain prototype vectors t_1^{proto} and t_2^{proto} are maintained by the RPD module for dual target domains, respectively. The prototypes are dynamically updated via EMA, ensuring they stably represent the feature distribution of the normal data in the target domain, formulated as

$$t_k^{proto} \leftarrow \mu \cdot t_k^{proto} + (1-\mu) \cdot \frac{1}{N_k} \sum_{i=1}^{N_k} f_{k,i} \tag{17}$$

where $k \in \{1,2\}$ corresponds to the dual target domains respectively. μ is the momentum factor, controlling the weight of historical prototypes. N_k denotes the number of samples in the current training batch for target domain k. $f_{k,i} \in \mathbb{R}^D$ is the fusion feature of the ith sample in the target domain k. Prototype updates are performed only during the training phase.

To drive the normal features of the two target domains toward each other, the RPD loss is introduced. Cosine distance is employed to quantify the discrepancy between a target feature and the opposite domain's prototype, enforcing a reciprocal drift that compresses the inter-target gap, which can be quantified as

$$\mathcal{L}_{RPD} = \frac{1}{N_1} \sum_{i=1}^{N_1} d_{\cos}\left(f_{1,i}, t_2^{proto}\right) + \frac{1}{N_2} \sum_{i=1}^{N_2} d_{\cos}\left(f_{2,i}, t_1^{proto}\right) \tag{18}$$

where the cosine distance $d_{\cos}(\cdot,\cdot)$ is defined as $d_{\cos}(a,b) = 1 - (a^{\mathrm{T}}b)/(\|a\|_2 \cdot \|b\|_2)$.

Therefore, minimizing the $\mathcal{L}_{RPD}$ loss is equivalent to maximizing the cosine similarity between the target-domain features and the prototype of the opposite target domain, which can be expressed as follows:

$$\max \cos(f_{1,i}, t_2{}^{proto}), \max \cos(f_{2,i}, t_1{}^{proto}) \tag{19}$$

According to the relationship between cosine similarity and angular distance $\cos(a,b) = \cos(\alpha)$, where α denotes the angle between two vectors. Consequently, minimizing $\mathcal{L}_{RPD}$ continuously reduces $\alpha(f_{1,i}, t_2{}^{proto})$ and $\alpha(f_{2,i}, t_1{}^{proto})$, indicating that features from each target domain are progressively attracted toward the prototype of the opposite domain.

Meanwhile, (17) can be recursively expanded as

$$t_k{}^{proto} = (1-\mu) \sum_{m=0}^{\infty} \mu^m \bar{f}_k^{(m)} \tag{20}$$

where $\bar{f}_k^{(m)} = \frac{1}{N_k} \sum_{i=1}^{N_k} f_{k,i}{}^{(m)}$ denotes the mean feature of the mth iteration.

Thus, $t_k{}^{proto}$ can be regarded as a smoothed estimate of the corresponding target-domain distribution center. As optimization proceeds, reducing $\alpha(f_{1,i}, t_2{}^{proto})$ and $\alpha(f_{2,i}, t_1{}^{proto})$ encourages the two target-domain distributions to move toward each other, thereby decreasing the discrepancy between their prototypes, as measured by

$$\alpha(t_1{}^{proto}, t_2{}^{proto}) \downarrow \tag{21}$$

For arbitrary features $f_1 \in T_1$ and $f_2 \in T_2$, the angular distance satisfies

$$\alpha(f_1, f_2) \le \alpha(f_1, t_2{}^{proto}) + \alpha(t_2{}^{proto}, t_1{}^{proto}) + \alpha(t_1{}^{proto}, f_2) \tag{22}$$

Since all three terms on the right-hand side are progressively minimized during optimization, the upper bound of $\alpha(f_1, f_2)$ is continuously tightened. As a result, $\alpha(f_1, f_2) \downarrow$, indicating that the discrepancy between the feature distributions of the two target domains is gradually compressed. The process is as follows:

$$\mathcal{L}_{RPD} \downarrow \Rightarrow \alpha(f_1, f_2) \downarrow \Rightarrow d(T_1, T_2) \downarrow \tag{23}$$

In summary, minimizing the RPD objective establishes a reciprocal attraction mechanism between dual target domains, which continuously reduces inter-domain feature discrepancy and drives the two target-domain distributions toward a shared embedding space. This provides a theoretical explanation for the feature alignment capability of the proposed RPD mechanism.

During training, a multi-task learning framework is utilized to jointly optimize both detection accuracy and cross-domain generalization. The overall objective function integrates several loss terms that collectively facilitate domain adaptation. In particular, a classification loss is applied to the labeled source domain to maintain attack detection performance on the source terrain, which is formulated as

$$\mathcal{L}_{Cls} = -\frac{1}{N_s} \sum_{i=1}^{N_s} \sum_{c=1}^{C} y_{s,i,c} \cdot \log(p_{s,i,c}) \tag{24}$$

where N_s denotes the number of samples in the current training batch of the source domain. C represents the number of attack categories and $y_{s,i,c} \in \{0,1\}$ is the true label of the ith sample in the source domain. Domain discrimination loss is designed based on GRL. By forcing domain invariance on the fused features through adversarial training, this module serves as the core mechanism for aligning the distributions of the source and target domains. The calculation is as follows:

$$\mathcal{L}_{Dom} = \mathcal{L}_S + \mathcal{L}_{T1} + \mathcal{L}_{T2} \tag{25}$$

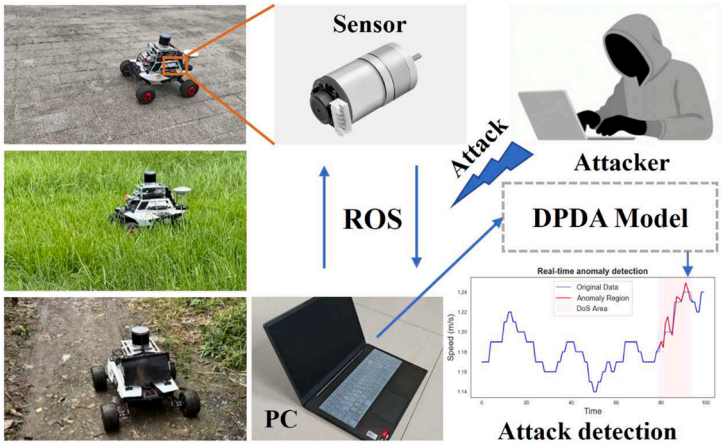


Fig. 3. Schematic diagram of the experimental procedure.

where the discriminative loss for each domain is a three-category cross-entropy, as exemplified by the source domain discriminative loss, taking the form of

$$\mathcal{L}_S = -\frac{1}{N_s}\sum_{i=1}^{N_s}\sum_{d=0}^{2} z_{s,i,d}\cdot\log(q_{s,i,d}) \tag{26}$$

Similarly, the forms of $\mathcal{L}_{T1}$ and $\mathcal{L}_{T2}$ are consistent with $\mathcal{L}_S$. The total loss is the weighted sum of the aforementioned losses, achieving a balance between attack detection accuracy and cross-domain generalization capability, defined as

$$\mathcal{L}_{Total} = \mathcal{L}_{Cls} + \mathcal{L}_{Dom} + \lambda\cdot\mathcal{L}_{RPD} \tag{27}$$

where λ is used to balances the contribution of the three loss terms.

During training, we employ a synchronous input strategy that feeds data from all three domains in each iteration. The corresponding losses are computed and aggregated within a single training step. The joint update prevents over-fitting that typically appears in staged training, exposes the model to both target distributions in each epoch, and accelerates domain adaptation convergence. As a result, source accuracy is preserved while real-time detection performance on the other terrains is markedly improved.

3. Experiments

In real-world scenarios, mobile robots operate in diverse road conditions. In this section, we select three typical terrains to form a gradient from simple to complex environments. Specifically, flat terrain is taken as the source domain, while grassy and sandy terrains are regarded as target domains. The experiment focuses on unidirectional domain adaptation from flat to grassy and sandy terrains, simulating generalization from structured to unstructured environments. The effectiveness of the proposed method is validated using data collected from real mobile robots across these different terrains. Extensive comparisons with single-domain detectors and domain adaptation baselines are first presented, followed by ablation studies that quantify the individual contribution of each DPDA module.

3.1. Experiment setup

3.1.1. Experimental procedure

The schematic diagram of cross-terrain real-time attack detection is given in Fig. 3. The experimental platform comprises a SCOUT MINI mobile robot, a speed sensor, and a laptop. Tests were conducted on flat terrain, grassy terrain and sandy terrain. The robot was driven autonomously along a predefined trajectory at a regular speed, while the speed sensor transmitted wheel speed data to the laptop via robot operating system (ROS) at 100 Hz. The DPDA model, deployed on the laptop, performed real-time analysis on the incoming data to detect potential anomalies. To emulate realistic cyber-attack scenarios, simulated attack signals were randomly injected into the velocity channel. Following established practices in related work (Wang et al., 2023; Wu et al., 2023; Zhang et al., 2024). Six types of attacks are considered: instant, constant, gradual drift, bias, denial of service (DoS), and hijacking attacks. Fig. 3 illustrates the system's response to a DoS burst. The DPDA model triggered an alarm within 1 second, prompting the laptop to issue a stop command through ROS, which brought the robot to a standstill.

3.1.2. Datasets

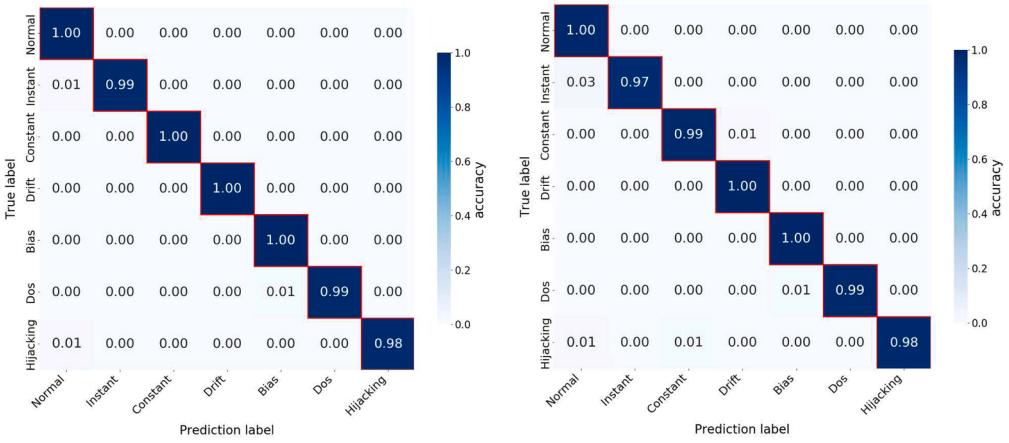
Two distinct datasets were utilized to ensure a comprehensive experimental evaluation. The first dataset was collected using our mobile robot platform across three terrain types: flat terrain (FT), grassy terrain (GT), and sandy terrain (ST). For ease of reference, we designate this dataset as the Cross-Terrain Attack Detection (CTAD) dataset, as it provides essential data support for the research. The second dataset is sourced from the Safe Pilot Model Deployment (SPMD) program's research data exchange. The dataset comprises driving records from over 2500 vehicles collected over two years and is widely recognized in the mobile robot safety domain. The SPMD dataset was employed to validate the generalization ability of the proposed method across diverse data domains, thereby enhancing the reliability of the experimental results.

3.1.3. Implementation details

During the domain adaptation phase, the model was trained for 80 epochs with a batch size of 64. The Adam optimizer was employed with a scheduled learning rate, initialized at 1e-3 and decayed by a factor of 0.96 every 1000 steps. The domain adaptation weight α in the GRL was set to 0.5. For the RPD module, the prototype update momentum coefficient μ was 0.99, and its loss weight was 0.3. Throughout training, every mini-batch contained an equal number of source and dual target instances to ensure simultaneous updates from all three domains.

3.2. Comparison under cross-terrain scenario

To validate the effectiveness of the proposed DPDA, we compare it with three single-domain anomaly detectors (DSA-CNN (Zhang et al., 2024), TSAN (He et al., 2025), SGOIDS Wei et al., 2025) and several representative domain adaptation algorithms (DFUDA (Lu et al., 2025), DACAD (Darban et al., 2025), E-MAPU (Ragab et al., 2026), DANN (Ghifary et al., 2014), DAN (Long et al., 2015), DAAN (Yu et al., 2019), DSAN (Zhu et al., 2020), SPA (Xiao et al., 2023), DDA-MLIC (abbreviated as D-M) Singh et al., 2024). The experimental results are summarized in Tables 2 and 3, with corresponding confusion matrices visualized in



(a) FT to GT. (b) FT to ST.

Fig. 4. Classification confusion matrix for DPDA.

Table 2
Detection performance of single-domain models.

Method	FT				FT to GT				FT to ST			
	Acc	Pre	Sens	F1	Acc	Pre	Sens	F1	Acc	Pre	Sens	F1
DSA-CNN (Zhang et al., 2024)	0.970	0.969	0.941	0.954	0.913	0.903	0.864	0.880	0.906	0.897	0.868	0.879
TSAN (He et al., 2025)	0.976	0.966	0.958	0.961	0.920	0.911	0.922	0.918	0.927	0.900	0.919	0.909
SGOIDS (Wei et al., 2025)	0.986	0.974	0.961	0.967	0.914	0.888	0.856	0.862	0.899	0.865	0.834	0.840
DPDA	**0.997**	**0.997**	**0.994**	**0.996**	**0.987**	**0.980**	**0.990**	**0.984**	**0.983**	**0.974**	**0.985**	**0.980**

Table 3
Detection performance of domain adaptation models.

Method	FT to GT				FT to ST			
	Acc	Pre	Sens	F1	Acc	Pre	Sens	F1
DANN (Ghifary et al., 2014)	0.947	0.928	0.924	0.922	0.932	0.910	0.915	0.910
DAN (Long et al., 2015)	0.958	0.940	0.950	0.943	0.946	0.931	0.943	0.933
DAAN (Yu et al., 2019)	0.957	0.941	0.941	0.941	0.955	0.938	0.942	0.940
DSAN (Zhu et al., 2020)	0.952	0.948	0.945	0.946	0.951	0.945	0.940	0.943
SPA (Xiao et al., 2023)	0.959	0.943	0.946	0.944	0.954	0.932	0.955	0.943
D-M (Singh et al., 2024)	0.974	0.968	0.967	0.967	0.973	0.966	0.963	0.964
DFUDA (Lu et al., 2025)	0.967	0.954	0.955	0.955	0.957	0.949	0.952	0.951
DACAD (Darban et al., 2025)	0.968	0.957	0.952	0.954	0.963	0.952	0.944	0.948
E-MAPU (Ragab et al., 2026)	0.974	0.970	0.973	0.969	0.971	0.961	0.968	0.965
DPDA	**0.987**	**0.980**	**0.990**	**0.984**	**0.983**	**0.974**	**0.985**	**0.980**

Fig. 4. As shown in the confusion matrix, all attack types are identified with an accuracy rate of nearly 100%, with only a small number of misclassifications occurring in a few categories, demonstrating excellent classification performance.

For the single-domain methods, a model trained exclusively on FT data is saved and directly evaluated on GT and ST data. As summarized in Table 2, all methods achieve high detection accuracy on the FT data, demonstrating strong within-domain capability. However, in the cross-domain scenarios, the performance of these single-domain methods drops sharply, revealing high sensitivity to terrain-induced distributional shift and limited generalizability. By relying solely on source-domain statistical features, these methods fail to adapt to the distinct distribution discrepancies present in the target domains. In contrast, high detection stability is maintained by the proposed DPDA in cross-domain settings, and the efficacy of its domain adaptation mechanism in mitigating inter-domain distribution discrepancies is underscored.

A systematic comparison with classical domain adaptation algorithms is presented in Table 3. Consistent and stable improvements are achieved by DPDA across all metrics. Traditional methods like DANN and DSAN struggle with terrain shift and fail to distinguish attack patterns. Even better-performing methods like E-MAPU and D-M inadequately capture anomalies in grassy terrain. DPDA achieves superior performance through effective domain alignment and preservation of key discriminative features. In the more challenging FT to ST adaptation scenario, the performance of all baseline methods is further constrained. Nevertheless, DPDA maintains a leading position, achieving an F1-score of 0.980 and reaffirming its strong adaptability in the presence of pronounced distribution shifts. The superior results validate the exceptional domain adaptation capacity of DPDA and its robustness against terrain-induced variations in attack features.

To verify whether the performance gains are statistically significant, paired t-tests are conducted between DPDA and several competitive baselines. Ten independent experimental runs are performed for each model under both cross-terrain adaptation tasks, with the significance level set to $\alpha = 0.05$. As reported in Table 4, DPDA exhibits substantially lower performance variability than all baseline methods. Furthermore, all comparisons yielded $p < .02$, with most $p < .01$, indicating that the gains are stable and statistically meaningful rather than caused by random variation.

3.3. Comparison under cross-dataset scenario

In real-world deployment of mobile robots, distribution shift can also arise from systematic differences such as varying data collection devices

Table 4
Statistical comparison results with paired t-test.

Method	FT to GT			FT to ST		
	Mean ± SD	t-statistic	p-value	Mean ± SD	t-statistic	p-value
D-M (Singh et al., 2024)	0.972 ± 0.012	3.029	0.014	0.970 ± 0.014	3.340	0.009
DFUDA (Lu et al., 2025)	0.967 ± 0.015	3.456	0.007	0.957 ± 0.017	4.638	0.001
DACAD (Darban et al., 2025)	0.968 ± 0.014	3.615	0.006	0.963 ± 0.014	4.035	0.003
E-MAPU (Ragab et al., 2026)	0.974 ± 0.011	3.138	0.012	0.971 ± 0.012	3.393	0.008
DPDA	0.987 ± 0.003	–	–	0.983 ± 0.004	–	–

Table 5
Detection performance of cross-dataset.

Method	CTAD to SPMD				SPMD to CTAD			
	Acc	Pre	Sens	F1	Acc	Pre	Sens	F1
DSA-CNN (Zhang et al., 2024)	0.912	0.916	0.878	0.893	0.889	0.906	0.882	0.887
TSAN (He et al., 2025)	0.926	0.931	0.902	0.910	0.900	0.922	0.911	0.903
SGOIDS (Wei et al., 2025)	0.906	0.915	0.876	0.880	0.894	0.903	0.871	0.876
DANN (Ghifary et al., 2014)	0.937	0.936	0.921	0.928	0.940	0.932	0.921	0.927
DAN (Long et al., 2015)	0.946	0.940	0.924	0.930	0.942	0.936	0.937	0.927
DAAN (Yu et al., 2019)	0.948	0.936	0.938	0.937	0.946	0.929	0.935	0.931
DSAN (Zhu et al., 2020)	0.945	0.938	0.934	0.936	0.948	0.935	0.932	0.933
SPA (Xiao et al., 2023)	0.953	0.942	0.947	0.944	0.951	0.936	0.948	0.941
D-M (Singh et al., 2024)	0.962	0.956	0.951	0.953	0.960	0.955	0.947	0.951
DFUDA (Lu et al., 2025)	0.952	0.943	0.942	0.942	0.953	0.939	0.941	0.940
DACAD (Darban et al., 2025)	0.958	0.949	0.943	0.946	0.955	0.943	0.946	0.944
E-MAPU (Ragab et al., 2026)	0.965	0.959	0.952	0.957	0.968	0.960	0.953	0.956
DPDA	**0.988**	**0.981**	**0.978**	**0.980**	**0.985**	**0.982**	**0.975**	**0.978**

Table 6
Performance comparison under different target domain settings.

Target Domain	RPD	FT to GT				FT to ST			
		Acc	Pre	Sens	F1	Acc	Pre	Sens	F1
Single	✗	0.926	0.914	0.928	0.920	0.916	0.904	0.919	0.911
Dual	✗	0.942	0.932	0.939	0.933	0.938	0.928	0.933	0.930
Dual	✓	0.987	0.980	0.990	0.984	0.983	0.974	0.985	0.980

and sensor parameters. Focusing solely on terrain differences may lead to performance degradation in scenarios involving the same terrain but different datasets. To address this issue, cross-dataset experiment is introduced to evaluate the generalization of the DPDA against distribution shift from heterogeneous data. The same attack generation strategy is applied to the SPMD dataset. The detection performance is summarized in Table 5, with both accuracy and F1-score further visualized in Fig. 5.

As observed, superior performance is demonstrated by the proposed DPDA compared with baseline methods. Specifically, in the CTAD to SPMD task, traditional domain adaptation methods achieve F1 scores mostly in the range of 0.928 to 0.944. Even the better-performing baselines, such as E-MAPU and D-M, show a clear gap compared with DPDA. Single-domain detectors, which ignore inter-dataset drift, suffer drastic degradation, underscoring the indispensability of adaptation. Similarly, in the SPMD to CTAD task, while all competing methods are constrained by inter-dataset distribution discrepancies, DPDA still leads with an accuracy of 0.985.

The superior performance of DPDA stems from its deep integration of the RPD module and dual-path framework. Temporal feature discrepancies between domains can be handled more finely, thereby mitigating performance degradation caused by domain shift. In contrast, algorithms such as DFUDA and DANN rely primarily on global domain alignment. While DACAD and DSAN, despite improvements in contrastive or local alignment, still fail to effectively constrain the feature drift between dual target domains. These results prove the effectiveness of DPDA in cross-terrain attack detection scenarios.

4. Model analysis and discussion

4.1. Justification of the dual-target setting

Different terrains exhibit distinct disturbance patterns, so features appearing in only one target domain are prone to terrain-specific noise, whereas features that remain consistent across multiple target domains are more likely to correspond to actual attack patterns. Therefore, it is necessary to align dual target domains simultaneously, as this enhances the model's ability to learn cross-terrain shared attack features. In addition, in real-world scenarios, mobile robots may frequently switch across terrains. If the model is retrained or separately adapted each time a new terrain is entered, temporal overhead and deployment latency will be incurred. Consequently, a model capable of simultaneous adaptation to dual target domains is of greater practical value.

To further validate the necessity of the dual-target domain setting, the detection performance of the model under single-domain and dual-domain scenarios is compared. Given that the RPD module is not required in the single-target domain scenario, the presence or absence of the RPD module is controlled to avoid simultaneous variations of the two variables. ✗ denotes the removal of the RPD module, whereas ✓ denotes its inclusion. The experimental results in Table 6 show that the dual-domain setting achieves better performance, indicating that more stable domain-invariant features can be learned through joint training with dual target domains. On this basis, further performance improvement is achieved after introducing RPD. This improvement can be attributed to the transformation of cross-target collaborative constraints

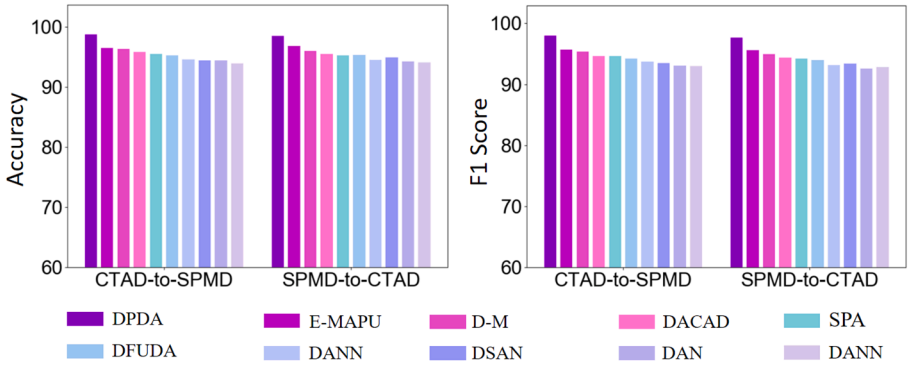
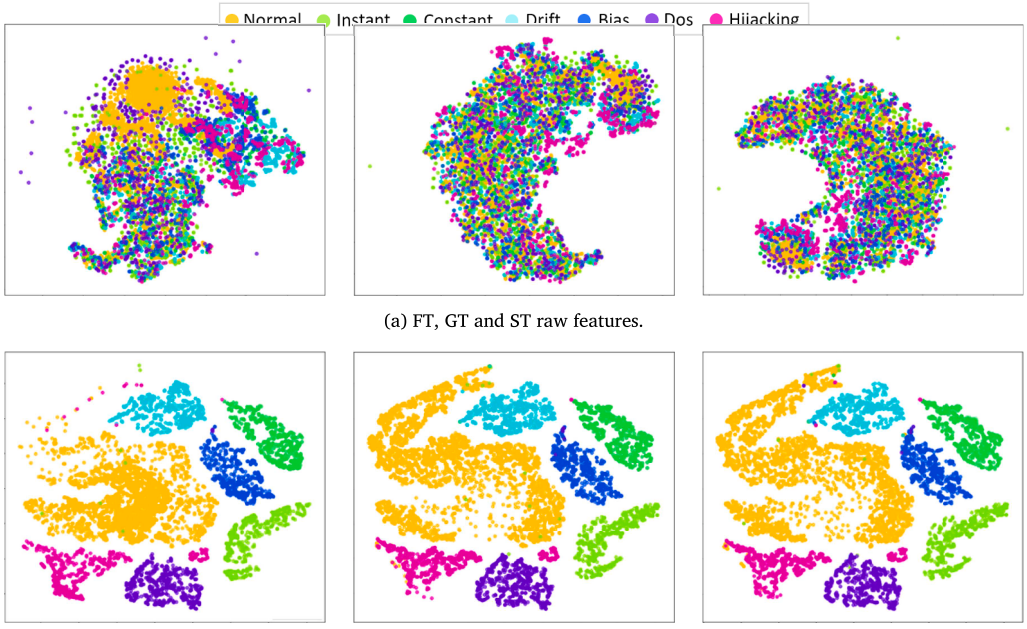


Fig. 5. Comparison of accuracy and F1 results cross-dataset scenario.



(a) FT, GT and ST raw features.

(b) FT, GT and ST features after classification.

Fig. 6. t-SNE visualization of features across three terrain types.

into an optimizable loss function, enabling terrain-specific noise to be suppressed and more intrinsic domain-invariant attack representations to be learned.

4.2. Empirical analysis

4.2.1. Feature visualization

The t-SNE visualizations of the feature space before and after applying DPDA are provided in Fig. 6. As shown in the t-SNE plots, without DPDA, feature points are highly intermixed and class boundaries are vague. The terrain-induced distribution shift between source and target domains substantially obscures the discrimination between normal and attacked samples. In contrast, after optimization with DPDA, the features of each category form compact and well-separated clusters. Furthermore, feature clusters of the same category across different domains exhibit high consistency in both distribution shape and relative positioning. The above results confirm that DPDA effectively aligns cross-domain feature distributions and preserves essential discriminative information, thereby achieving both accurate domain adaptation and reliable attack classification.

4.2.2. Distribution discrepancy

$\mathcal{A}$-distance serves as a classical metric for measuring global distribution discrepancy, which is approximated by computing the generalization error of a binary classifier trained to distinguish source and target domain samples. A smaller value indicates a higher degree of domain alignment. As illustrated in Fig. 7, the lowest $\mathcal{A}$-distance in both the FT to GT and FT to ST tasks are achieved by the proposed DPDA method. This result demonstrates that the distribution discrepancy between different terrains is effectively reduced by DPDA, and more domain-invariant features are learned for mobile robot anomaly detection in varied terrain environments. Notably, in the GT to ST task, the $\mathcal{A}$-distance for DPDA is substantially lower than that of all competing methods, which validates the effectiveness of the proposed RPD module. The distribution alignment of the two target domains is facilitated by this module through the mutual guidance of their prototype representations, thereby enhancing the cross-terrain generalization capability of the learned features.

4.3. Terrain separability analysis

Given that spectral features are highly dependent on terrain conditions, the model is susceptible to overfitting on terrain-specific artifacts. This implies that its discriminative ability may be driven by terrain-related cues rather than by the underlying attack patterns themselves. From the perspective of representation learning, such overfitting manifests as terrain-specific information retained in the feature space. If this information remains, samples from different terrains will be highly sep-

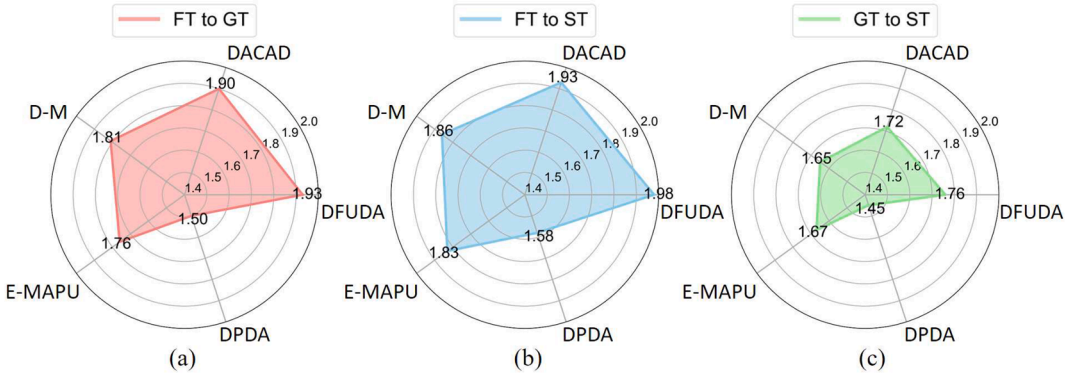


Fig. 7. $\mathcal{A}$-distance on tasks FT to GT, FT to ST and GT to ST.

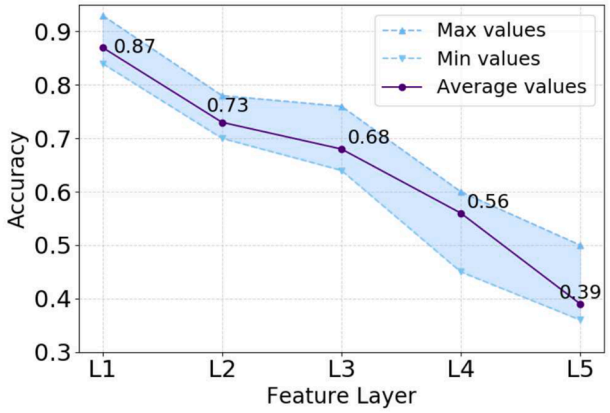


Fig. 8. Terrain classification accuracy across different feature layers of the model. L1: flattened features of raw sensor data. L2: features obtained by concatenating the outputs of the dual branches. L3: features processed by the TFC module. L4: features fused by the BCA module. L5: final features after RPD alignment.

arable and can be distinguished by a simple classifier. Based on this principle, terrain separability analysis is conducted on features extracted from different network layers. Specifically, DPDA is divided into five feature layers (L1-L5) from the input layer to the final output layer. During feature extraction, all network parameters are frozen to ensure that the learned feature representations are not updated. Subsequently, a linear SVM is employed for terrain classification on features from each layer. Since linear classifiers directly reflect the linear separability of terrain-related information in the feature space, the resulting classification accuracy can be utilized to evaluate the degree of residual terrain-specific information at different stages.

The results are shown in Fig. 8. As the features evolve layer by layer, the classification accuracy consistently decreases, indicating that DPDA progressively suppresses terrain-specific information through each module. In particular, the performance drop from L2 to L3 suggests that the TFC module filters terrain-related frequency components through dynamic gating. The further reduction from L4 to L5 is consistent with the role of RPD in reducing domain discrepancies through reciprocal alignment. Ultimately, the accuracy at L5 approaches random chance (0.33), indicating that terrain-discriminative information has been effectively suppressed from the learned representations. Despite the substantial reduction in terrain separability, DPDA still maintains strong anomaly detection performance, demonstrating that the model captures terrain-invariant attack patterns rather than terrain-dependent features.

4.4. Evaluation of real-time requirements

To assess the practical viability of deploying the proposed DPDA in real-world vehicular scenarios, a quantitative analysis of computational complexity and detection latency is conducted in this study. The experimental results are presented in Fig. 9, where the proposed DPDA model achieves a favorable balance between computational complexity and inference efficiency. The complexity of this model reaches approximately 50.8M FLOPs with a latency of 58 ms, which satisfies the response requirement of 10 to 100 ms for the highest priority vehicular safety services such as attack detection (Abualhoul et al., 2016). Although lower latency is exhibited by some comparative models, their detection accuracy is insufficient to meet practical safety requirements. By contrast, the DPDA model is shown to effectively control computational and temporal costs while maintaining competitive detection performance. The algorithm is deployed on a mobile robot for real-time anomaly detection. Fig. (10) presents the real-time anomaly detection of the DPDA-deployed mobile robot across varying terrains, with a demonstration video available at https://github.com/yuwanggg/anomaly-detection-demo.

4.5. Ablation study

To evaluate the contribution of key components in the proposed DPDA, we conduct ablation experiments. The detailed experimental protocol is as follows.

1) Network A: Remove the BCA mechanism.
2) Network B: Remove the TFC module.
3) Network C: Remove the RPD module.

The necessity of these three modules is validated by progressively ablating them. As shown in Table 7. The complete DPDA model achieves the best performance on both FT to GT and FT to ST tasks, outperforming all ablated variants across all metrics. Specifically, removing the RPD module resulted in the most significant performance drop, followed by the removal of the TFC module. While ablating the BCA mechanism also degrades performance, its impact is relatively minor. The findings indicate that all modules contribute positively to the model's effectiveness, with RPD and TFC being particularly critical for maintaining high performance. The observed differences in contribution further validate DPDA's design logic of hierarchical collaboration among multiple modules. The synergy among these modules effectively enhances feature dis-

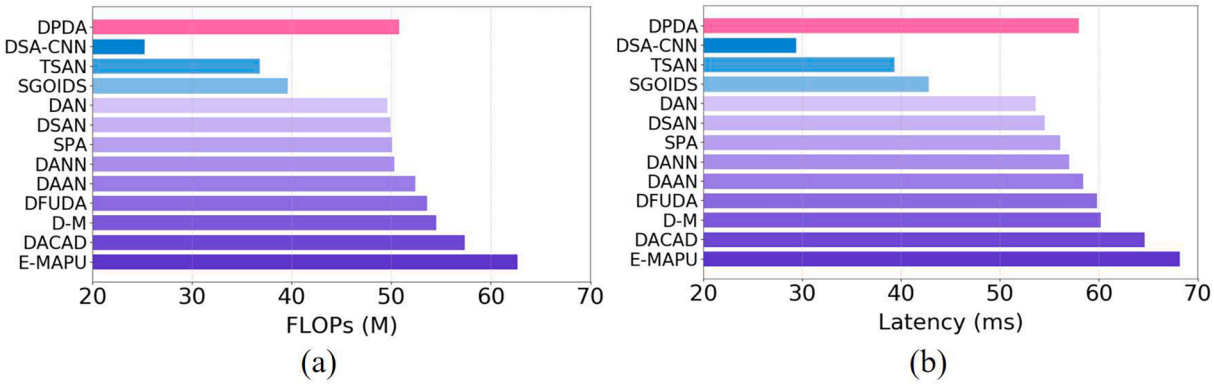


Fig. 9. (a) Computation complexity. (b) Detection latency.

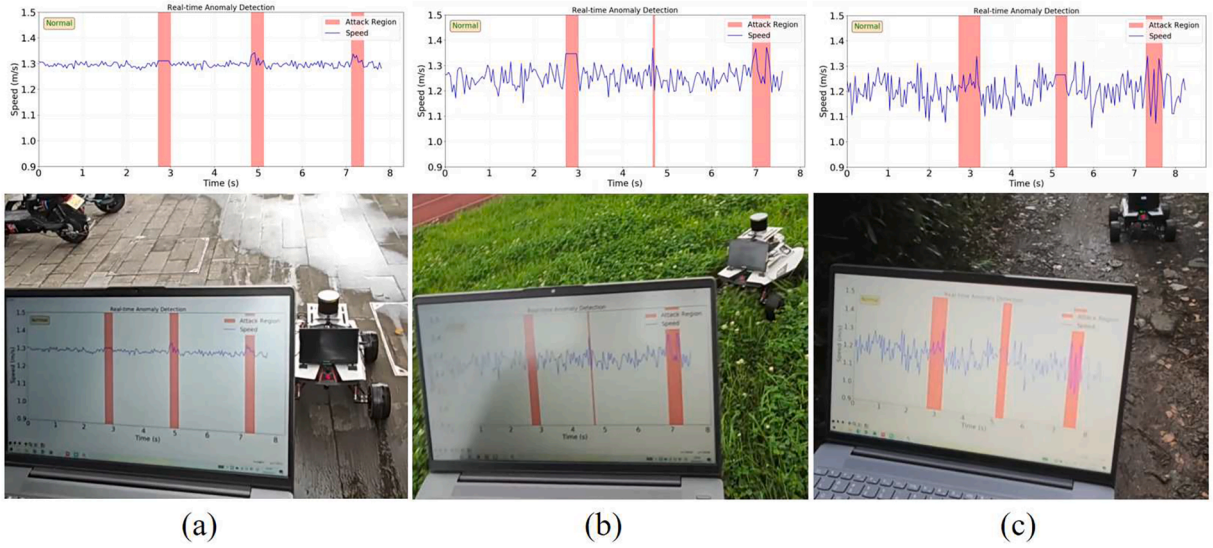


Fig. 10. Real-time anomaly detection of the mobile robot across varying terrains. (a) flat terrain. (b) grassy terrain. (c) sandy terrain.

Table 7
Performance of each network in the ablation experiment.

Method	FT to GT				FT to ST			
	Acc	Pre	Sens	F1	Acc	Pre	Sens	F1
Network A	0.979	0.962	0.970	0.966	0.974	0.961	0.968	0.963
Network B	0.956	0.947	0.939	0.941	0.954	0.947	0.937	0.940
Network C	0.942	0.932	0.939	0.933	0.938	0.928	0.933	0.930
DPDA	**0.987**	**0.980**	**0.990**	**0.984**	**0.983**	**0.974**	**0.985**	**0.980**

criminability and domain adaptability across diverse terrain scenarios, underscoring that each module is indispensable for the overall performance improvement.

5. Conclusion

In this paper, a novel domain adaptation method, namely DPDA, is proposed to address the challenge of cross-terrain attack detection in mobile robots. By extracting domain-invariant discriminative features, the system's capability for cross-terrain attack detection is significantly enhanced. The dual-path backbone employs temporal convolution and spectral analysis to capture dynamic attack behaviors from sensor data. Building on this, the TFC module and BCA mechanism are introduced to capture the stable features of attacks across different terrains. To narrow the distribution gap between dual target domains, the proposed RPD utilizes reciprocal guidance of dual target feature prototypes together with adversarial constraints, stabilizing the decision boundary across terrains. Extensive experiments are conducted on the real-world dataset collected by our mobile robot platform. The results convincingly demonstrate DPDA's superiority in tackling cross-terrain attack detection problems. In the future, we will explore domain adaptation strategies under more diverse terrains and attack scenarios, and pursue lightweight optimization to further satisfy real-time inference requirements.

CRediT authorship contribution statement

Yu Wang: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization; **Zhitao He:** Writing – original draft, Methodology, Data curation, Conceptualization; **Yongyi Chen:** Visualization, Supervision, Formal analysis, Data curation; **Dan Zhang:** Writing – review & editing, Supervision, Formal analysis; **Hamid Reza Karimi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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